

The Regulatory System

The human body is a cooperative of 38 trillion cells organized into different systems. Keeping them functioning at their best is a system of feedback mechanisms controlled by the brain.

Maintaining stability

The process of maintaining a stable internal environment is called homeostasis. Key functions, such as breathing, heart rate, pH, temperature, and ion balances have to be kept within strict operating limits to prevent us from becoming ill. As the body works, its systems are constantly being moved away from their balance or set point (the value at which a system works best). When the change becomes too great, the body initiates a feedback loop that returns the system to its ideal level. Many of these functions are controlled by a part of the brain stem called the reticular formation.

3 Signals forwarded

Signals are then sent directly to the thalamus and hypothalamus, as well as to the appropriate areas of the cerebral cortex for a decision and response to the stimulus.

Excitatory area of reticular formation amplifies important signals

WHAT IS THE RETICULAR FORMATION?

The reticular formation consists of more than 100 nuclei that project to the forebrain, cerebellum, and brain stem, controlling many of the body's vital functions.

Impulses travel up spinal cord

1 Signals travel up the spinal column

Incoming sensory signals from all over the body travel to the reticular formation.

2 Signals processed

In the reticular formation, unwanted signals are suppressed in the inhibitory area, while others are amplified in the excitatory area.

Inhibitory area of reticular formation dampens unwanted signals

GENERAL ANESTHETICS

A vital part of modern surgery, how general anesthetics work is not fully understood. What is known is that they act on the reticular activating system (comprising the reticular formation and its connections) to suppress awareness and on the hippocampus to temporarily suspend memory formation. Anesthetics also affect the nuclei of the thalamus, preventing the flow of sensory information from the body to the brain.



Signals travel to various areas of cerebral cortex

Hypothalamus regulates sleep, hunger, and body temperature

Thalamus relays sensory signals to cerebral cortex

THALAMUS

MEDULLA

SPINAL CORD